

Supplementary Materials for

**Sequential integration of multimodal data from serum improves the predictive performance of
hepatic lipid accumulation in mice.**

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Figures S1 to S4.

Table S1.

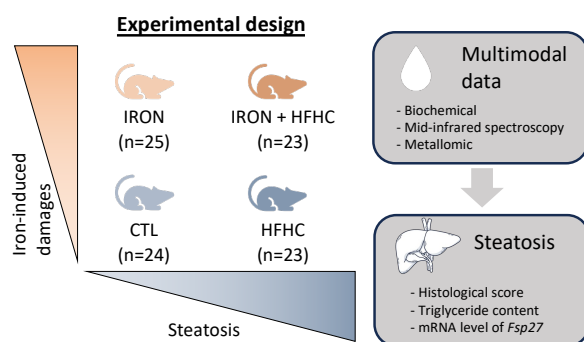


Figure S1: Experimental design. Two experimental variables were introduced: mice were subjected either to a control diet or to a high-fat, high-carbohydrate (HFHC) diet, and, independently, received injections of either dextran or iron-dextran. The steatosis (i.e., hepatic lipid accumulation) has been evaluated through 3 biological scales. We constructed several datasets. The first dataset consists of biochemical measurements from serum. The second dataset is derived from mid-infrared (MIR) spectroscopy performed on serum, providing molecular fingerprints specific to each mouse. The third dataset comprises quantitative measurements of metals and trace elements in the serum.

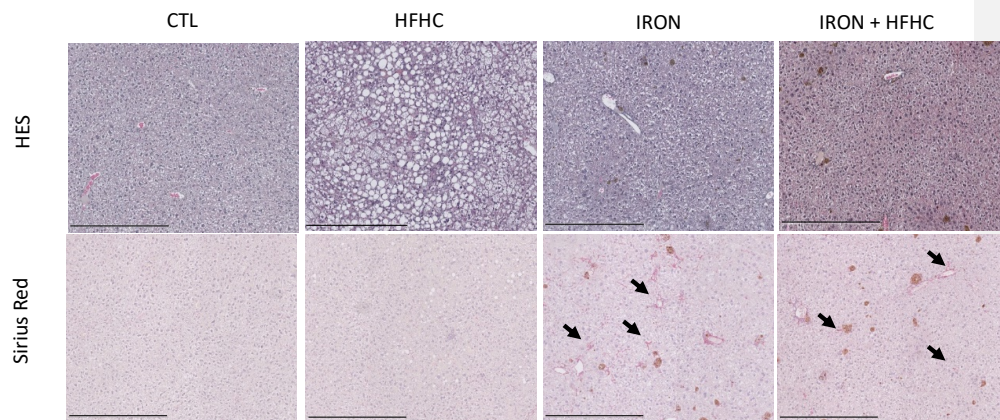


Figure S2. Liver staining. HES and Sirius Red staining of the liver of CTL, HFHC, IRON, and IRON/HFHC mice at 22 weeks. The scale bar corresponds to 500 μ m. Black arrows are related to collagen.

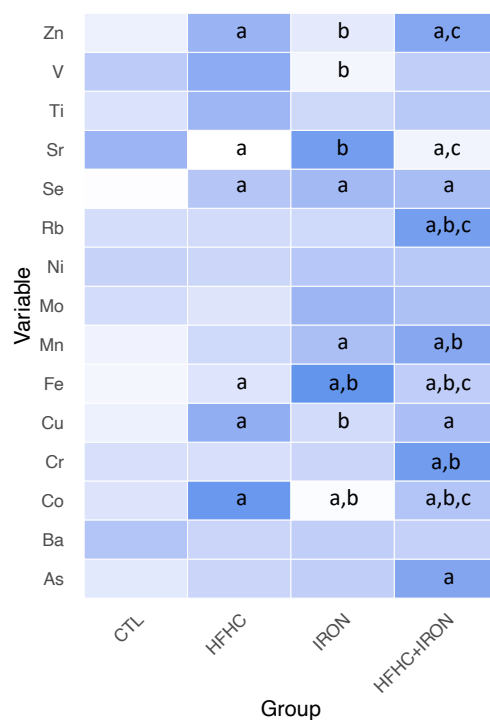
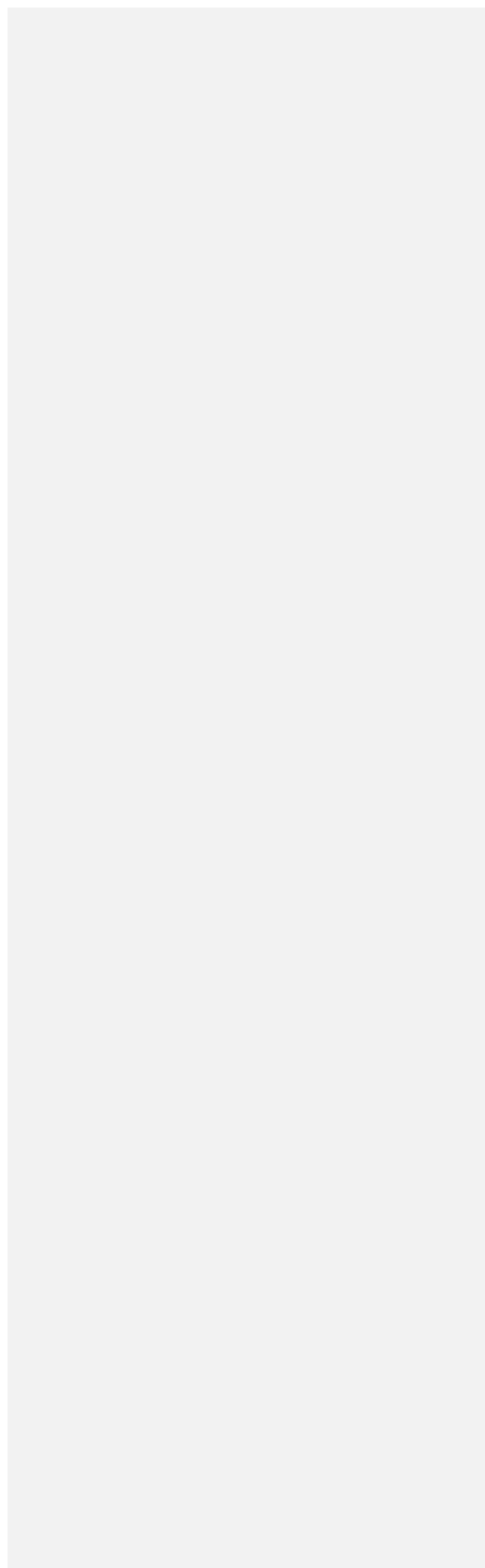


Figure S3. Heatmap of trace element levels across experimental groups. Increasing intensity of blue indicates higher values. Statistically significant differences are annotated as follows: *a* denotes a significant difference versus CTL, *b* versus HFHC, and *c* versus IRON. Statistical analysis was performed using multiple pairwise Wilcoxon tests with Bonferroni correction.

Commenté [MOU1]: Ajouter les correspondances des abréviations.

Figure S4. Representative spectra depending on experimental groups, with the important wavenumbers defining the lipid accumulation in the liver of mice.



5' -3'	Fwd	Re
<i>A2m</i>	GAAACAGAGATCAGCATGACCAGTA	CATTGGATTGTGGTAGAGGAACT
<i>Crp</i>	TGGTGGGAGACATCGGAGAT	GCCCGCCAGTTCAAAACAT
<i>Hprt1</i>	CCGGAATCCCTATCTTTAGTCC	GGGTCAGTCCAGTGCCATAAG
<i>Nrf2</i>	CGAGATATACGCAGGAGAGGTAAGA	GCTCGACAATGTTCTCCAGCTT

Table S1. Primers used for liver tissue. *Crp*, C - reactive protein; *Hprt1*, hypoxanthine phosphoribosyltransferase 1; *Nrf2*, Nuclear factor (erythroid-derived 2)-like 2; *A2m*, α 2-macroglobuline